BME453- COMPUTER AIDED MACHINE DRAWING-II LAB

Course Outcome (CO)		Bloom's Knowledge Level (KL)
	At the end of course , the student will be able to	
CO 1	Understand and apply 3D software to develop a part model	**K3**
CO 2	Understand conventional representation of machine components, and welded joints	**K2**
CO 3	Understand and apply the basis of fit or limit system	**K3**
CO 4	Understand about Plummer Block Bearing, Machine Vice, Screw Jack, Engine Stuffing box.	**K2**
CO 5	Create 3D part models and assemblies of various machine components	**K3**
	DETAILED SYLLABUS	**0-0-2**

Unit I

Introduction to Part Modelling: Introduction to part modelling of simple machine components using any 3D software (like CATIA, PRO E, UGNX, Autodesk Inventor or SOLIDWORKS) covering all commands/ features to develop a part model (Minimum 24 machine components need to be developed).

Unit II

Introduction: Conventional representation of machine components and materials, Conventional representation of surface finish, Roughness number symbol, Symbols of Machine elements and welded joints. Classification of Drawings: Machine drawings, Production drawing, part drawing and assembly drawing. Introduction to detail drawing and bill of materials (BOM).

Unit III

Limits, Fits and Tolerances: General aspects, Nominal size and basic dimensions, Definitions, Basis of fit or limit system, Systems of specifying tolerances, Designation of holes, Shafts and fits, commonly used holes and shafts. List of Standard Abbreviation used.

Unit IV

Part Modelling & Assemblies of: Plummer Block Bearing, Machine Vice, Screw Jack, Engine Stuffing box, Lathe Tailstock, Feed Check Valve and Rams Bottom Safety Valve.

Note: All drawing conforms to BIS Codes.

Books and References:

1. Textbook of Machine Drawing, K C John, PHI.
2. Machine Drawing by K.R. Gopalakrishna, Subhas Stores.
3. A Textbook of Machine Drawing by PS Gill from S.K. Kataria& Sons.
4. Machine Drawing-KL Narayana, P Kannaiah, KV Reddy, New Age publications.
5. Engineering Graphics with AutoCAD, Bethune, PHI.
6. Machine Drawing, N. Siddeshswar, P Kannaiah, VVS Shastry, Tata McGraw Hill.
7. Fundamentals of Machine Drawing, Dr Sadhu Singh & P L Shah, Prantice Hall India.
8. Autodesk Inventor by Examples, Sam Tikoo, Wiley.

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